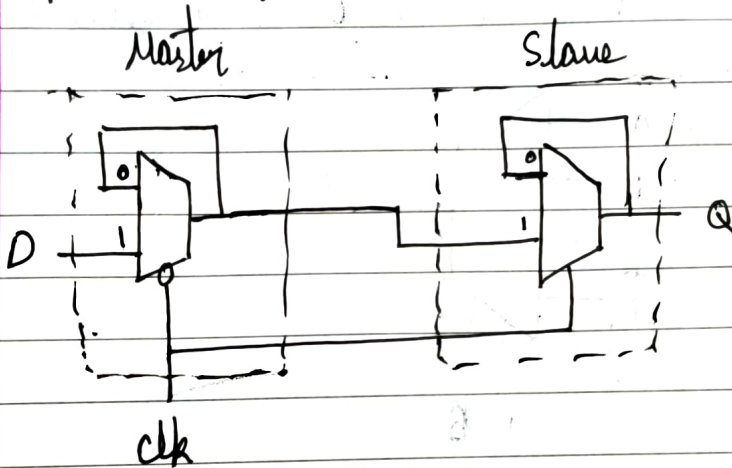


DAY 41.) Next State Table

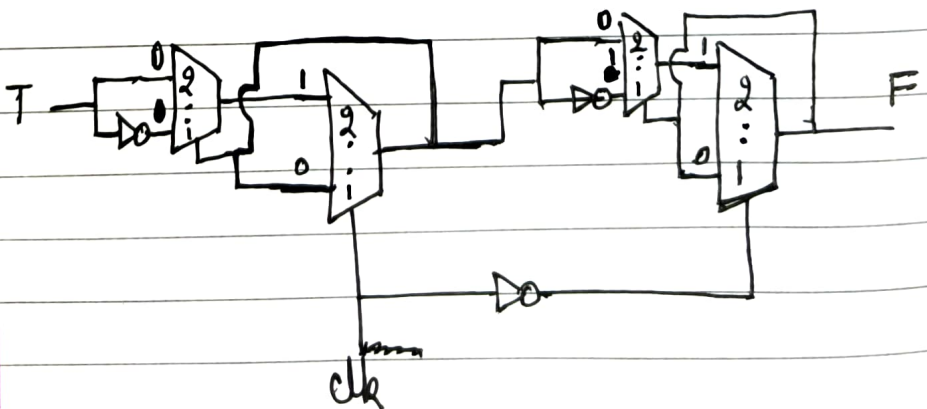
x	Q	Q ⁺
0	0	0
0	1	1
1	0	1
1	1	0

This table acts as similar to T-FF

2.) positive edge triggered D-FF using 2:1 MUX



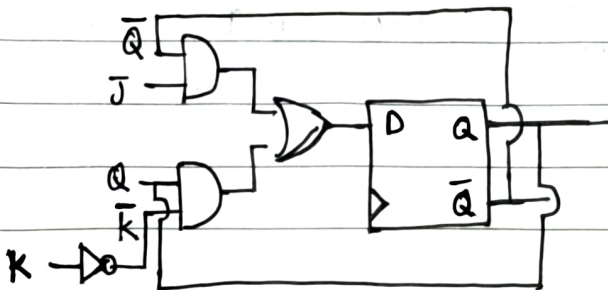
3.) negative edge triggered T-FF using 2:1 MUX



4.) a.) D FF $\Rightarrow$ JK

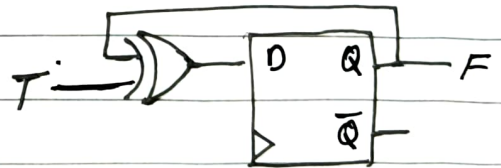
J	K	Q	Q ⁺	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

Using k map: $D = Q\bar{K} + \bar{Q}J$



b.) T $\Rightarrow$ D FF

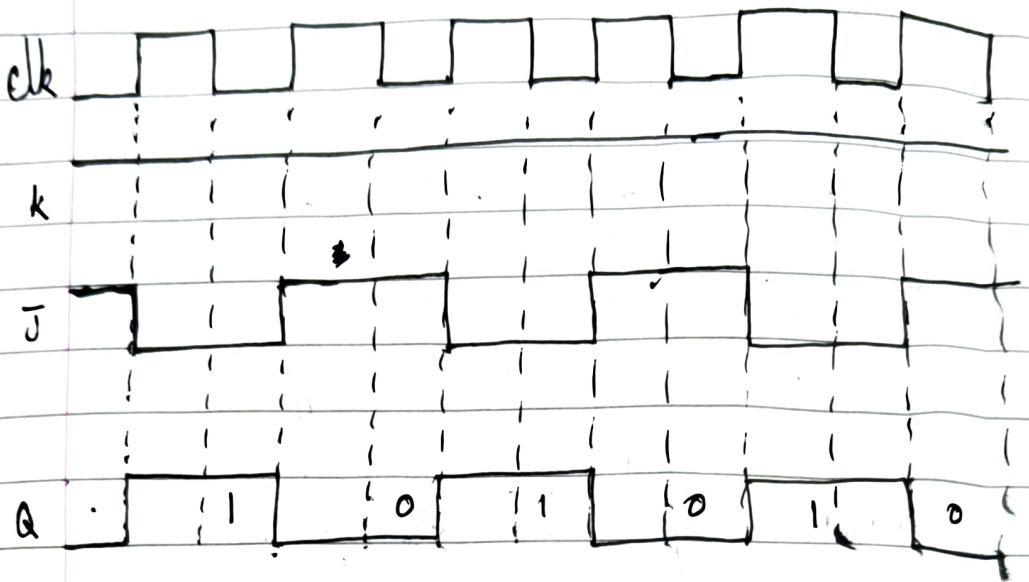
D	Q	Q ⁺	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0



Using kmap: $T = D\bar{Q} + \bar{D}Q$
 $= D \oplus Q$

5) Timing diagram

Assuming $Q = 0$, $\bar{Q} = 1$ initially



$\therefore$ output sequence would be 0, 1, 0, 1, 0